Analog Transmission & Bandwidth Utilization (Multiplexing)

Coverage: Digital Modulation (ASK, FSK, PSK, QAM), Multiplexing (FDM, WDM, Synchronous vs Statistical TDM), Spread Spectrum

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1. Modulation & QAM

• ASK, FSK, PSK (BPSK, QPSK). QAM modulates both amplitude and phase (16-QAM = 4 bits/symbol).

2. Multiplexing

• FDM (Frequency), WDM (Optical wavelength), Synchronous TDM (Round-robin slots), Statistical TDM (On-demand dynamic slots).

■ KTU University High-Yield Exam Questions

• Q: Explain QAM with constellation diagrams and compare Synchronous vs Statistical TDM. (8 Marks)